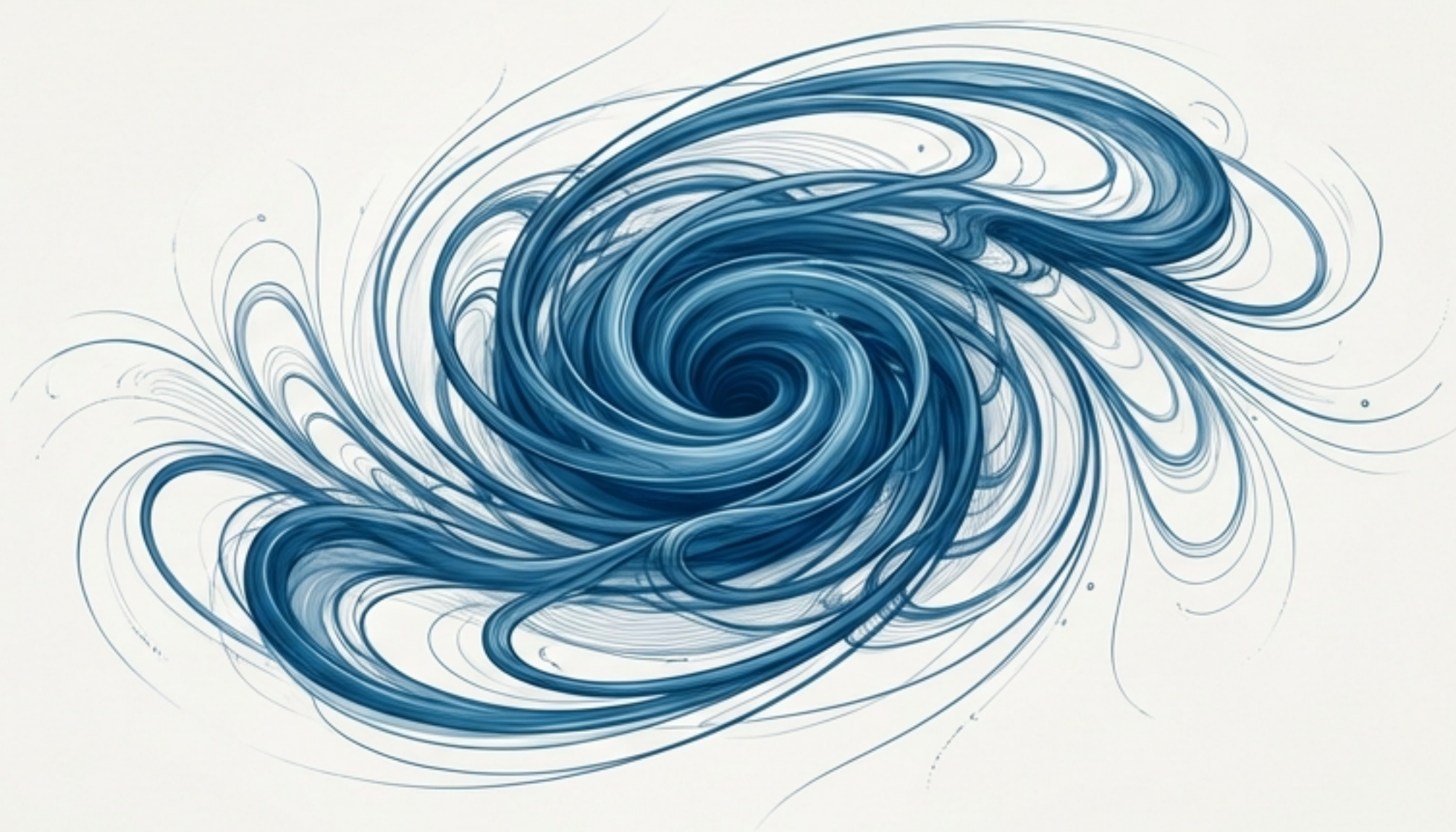


Time from Swirl

A Hydrodynamic Origin of Proper Time in Swirl-String Theory



This presentation develops a framework where the rate of time's passage for a particle is not governed by spacetime curvature, but by the rotational energy of its underlying structure—a 'swirl.' Proper time emerges as a local, dynamic quantity.

The SST Framework: A Condensate and Knotted Strings

The Medium

Inter

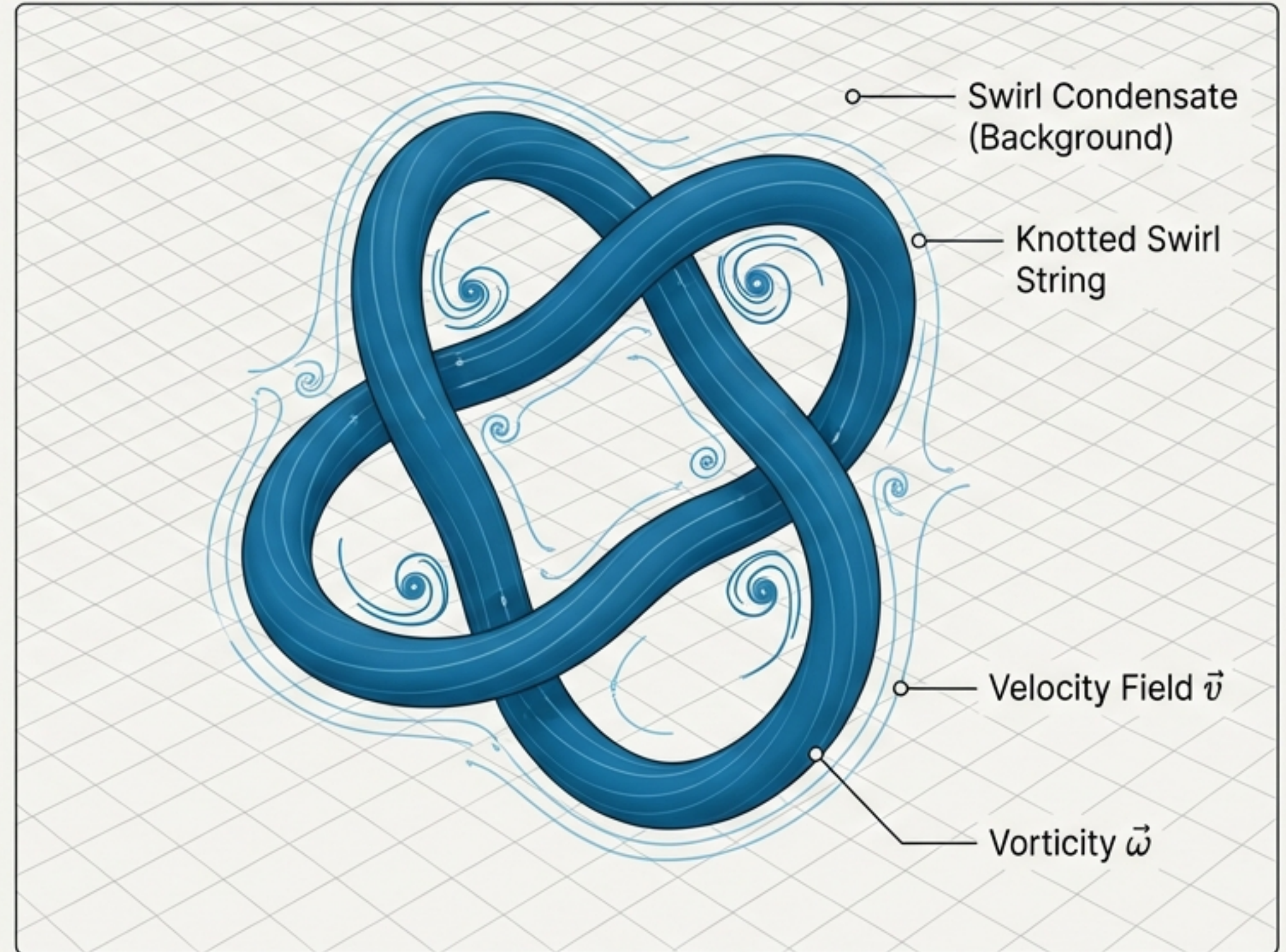
Physics is modeled on an incompressible, inviscid “swirl condensate” with an effective density, ρ_f . This is the background medium.

The Matter

Particles are modeled as knotted “swirl strings” embedded within this condensate.

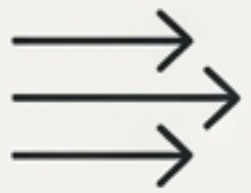
The Kinematics

The state of the condensate is described by a velocity field $\vec{v}(\vec{x}, t)$ and its corresponding vorticity, $\vec{\omega}(\vec{x}, t) = \nabla \times \vec{v}(\vec{x}, t)$.



Describing the Energy of a Swirl

$$H_{swirl}(\vec{x}) = \frac{1}{2}\rho_m \left(|\vec{v}(\vec{x})|^2 + \ell^2\omega|\vec{\omega}(\vec{x})|^2 \right)$$



Bulk Kinetic Energy

The standard energy of motion of the condensate.



Swirl Tension

A short-range term representing the energy stored in the local rotation and structure (structure (vorticity) of the fluid. Analogous to tension in a stretched string.

$\ell\omega$: A characteristic length scale on the order of the swirl's core radius, r_c .

The Swirl Clock: Linking Motion to Time

The internal evolution of a swirl-bound system slows down relative to an asymptotic observer's time (t_∞). This slowdown is dictated by the local tangential swirl speed.

Analogy: Special Relativity



$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$dt = \gamma dt_0$$

$\leftrightarrow$

$\leftrightarrow$

The Swirl Clock Factor, $S_\ominus$



$$dt(\vec{x}) = dt_\infty S_\ominus(t)(\vec{x})$$

$$S_\ominus(t)(\vec{x}) := \sqrt{1 - \frac{|\mathbf{v}_\perp(\vec{x})|^2}{\|\mathbf{v}_\ominus\|^2}}$$

Calculating the Clock Rate at the Swirl Core

The Model

We use a standard axisymmetric profile for the tangential velocity:
 $v_{\theta}(r) = \|\mathbf{v}_{\Theta}\| \left(\frac{r}{r_c}\right) e^{-r/r_c}$. This velocity peaks at the core radius, $r = r_c$.

The Derivation

1. Substitute the velocity profile into the Swirl Clock definition:

$$S_{\Theta}(t)(r) = \sqrt{1 - \left(\frac{r}{r_c}\right)^2 e^{-2r/r_c}}$$

2. Evaluate at the core radius $r = r_c$, where the effect is strongest:

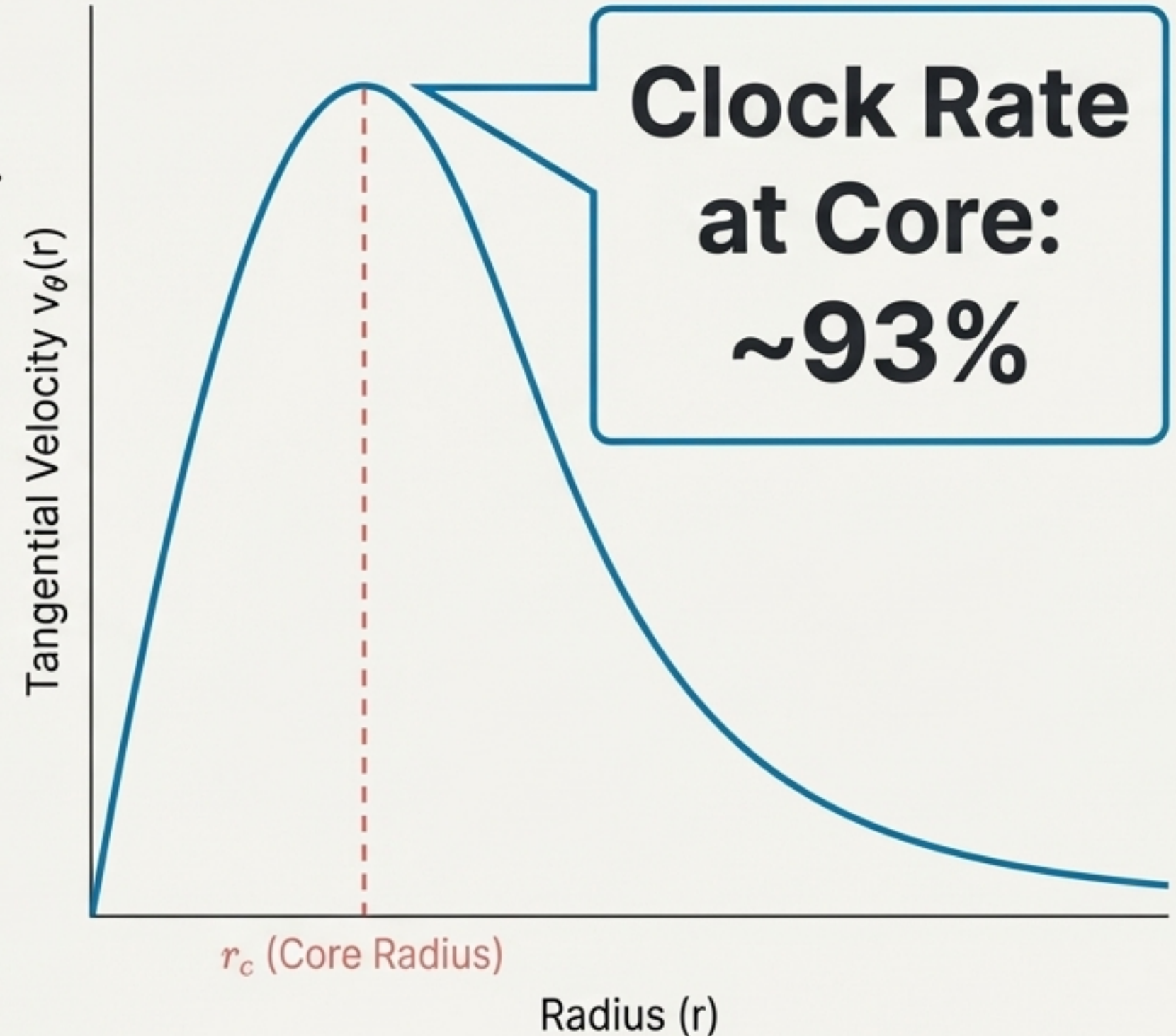
$$S_{\Theta}(t)(r_c) = \sqrt{1 - e^{-2}}$$

The Result

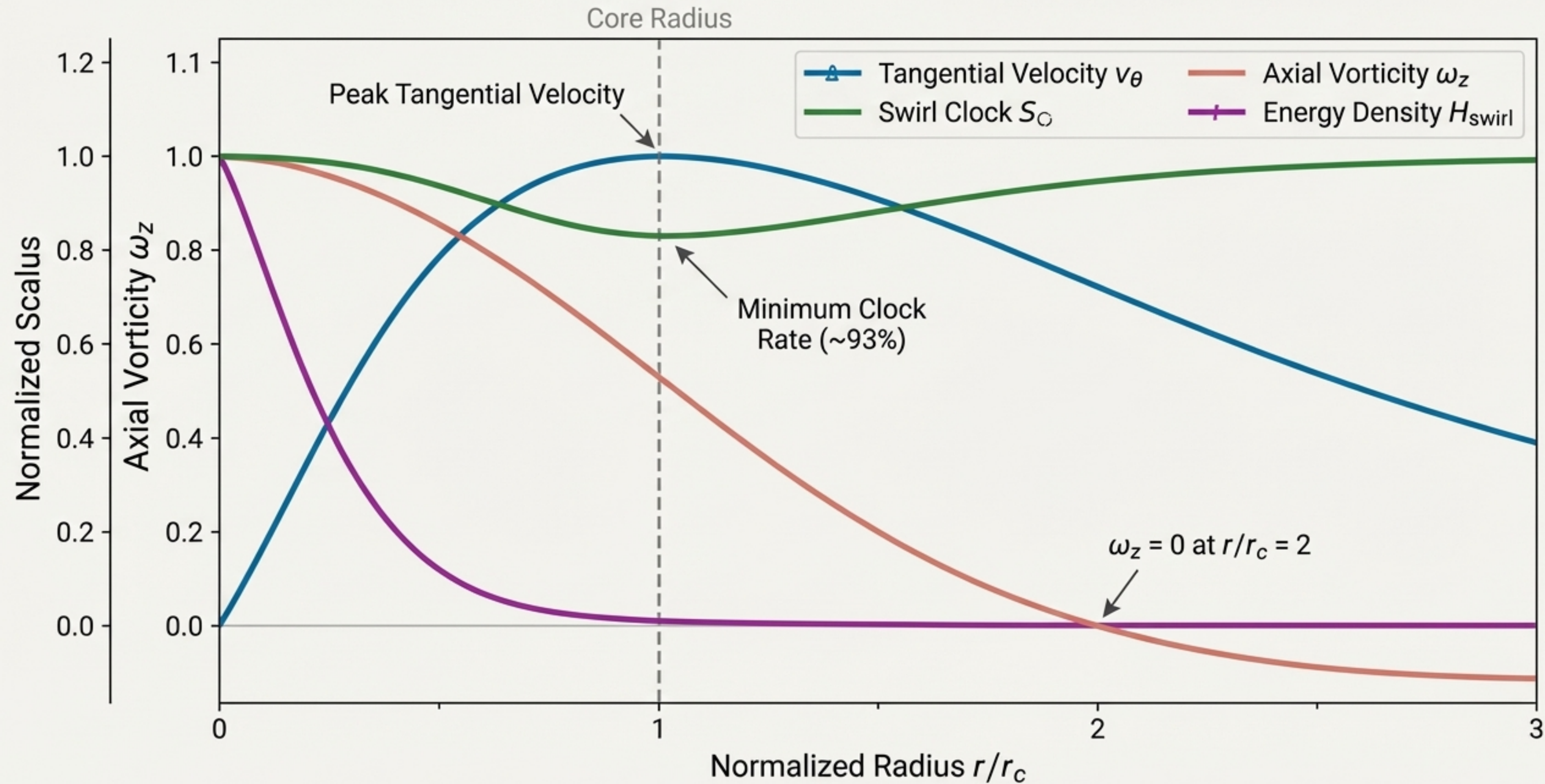
$$S_{\Theta}(t)(r_c) \approx 0.92987$$

Takeaway

At its core, the swirl's internal clock runs at approximately 93% of the rate of a distant observer's clock.



Anatomy of a Canonical Swirl String



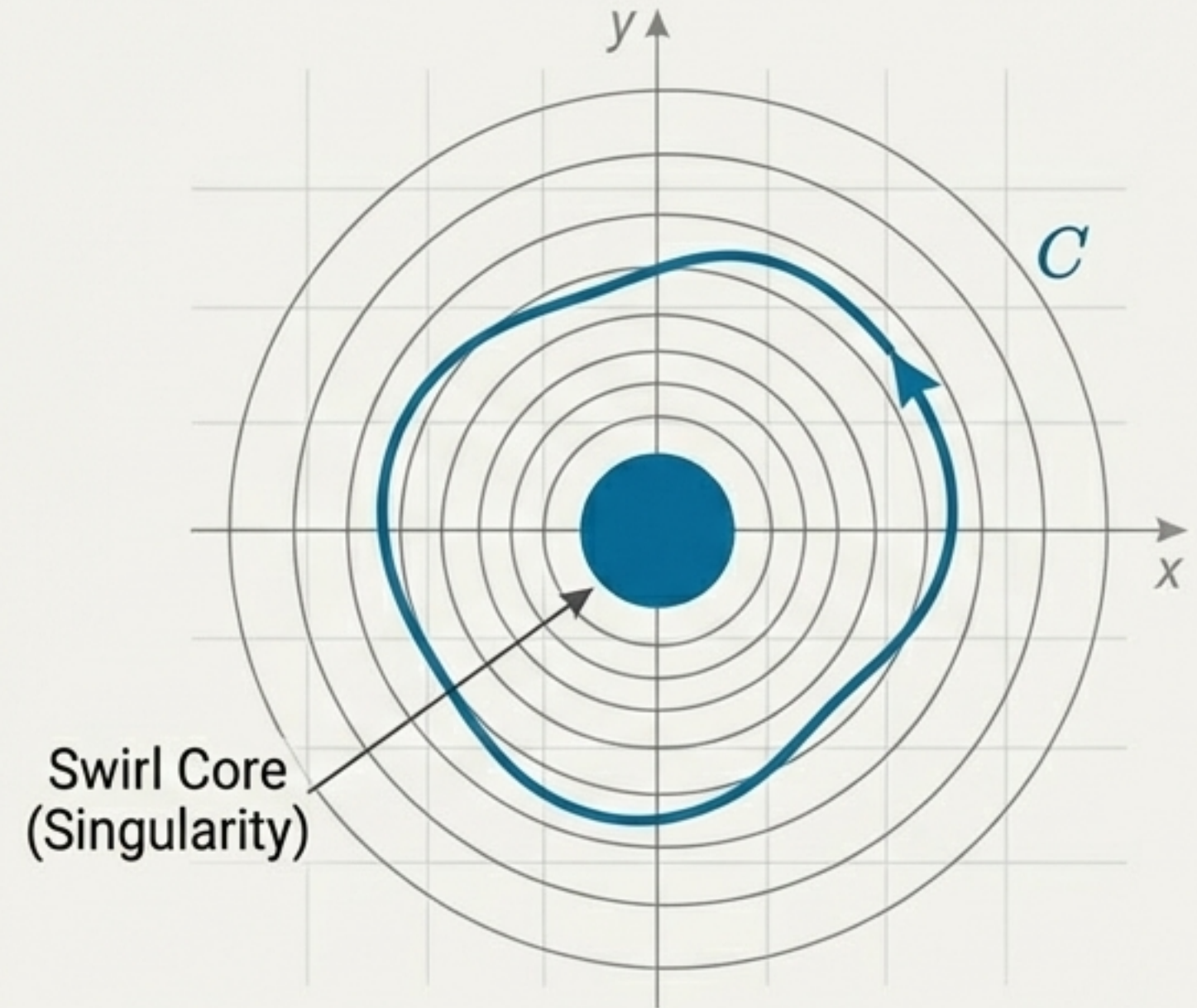
A Deeper Structure: Complex Analysis and Circulation

The Connection

In the 2D plane transverse to a straight swirl string, the flow is incompressible and (outside the core) irrotational. This allows the use of complex analysis.

Key Tools

- Complex Coordinate: $z = x + iy$
- Complex Potential: $W(z) = \Phi + i\Psi$
- Complex Velocity: $U(z) = v_x - iv_y = \frac{dW}{dz}$



Circulation, Γ_C

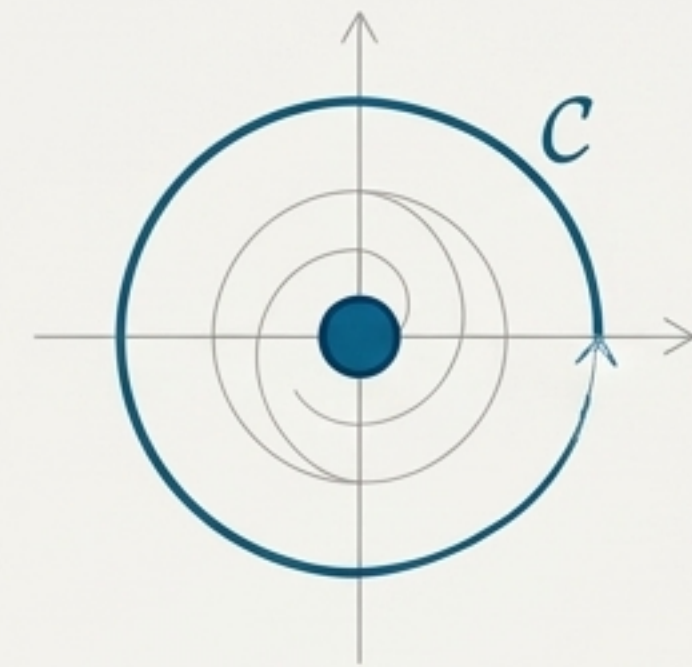
The line integral of velocity around a closed loop C . It measures the total "amount of swirl" enclosed by the loop.

$$\Gamma_C = \oint_C \vec{v} \cdot d\vec{\ell} = -\Im \oint_C U(z) dz$$

The Topological Leap: Circulation is Quantized

The Logic (via Cauchy's Integral Theorem)

1. If a loop C does **not** enclose the swirl core, $U(z)$ is analytic inside, so $\oint U(z) dz = 0$, which implies $\Gamma_C = 0$.
2. If the loop **does** enclose the core (a singularity), the integral is non-zero:
$$\oint U(z) dz = 2\pi i \times \text{Res}(U, z = 0).$$



Loop Outside Core

$$\Gamma_C = 0$$

The Result

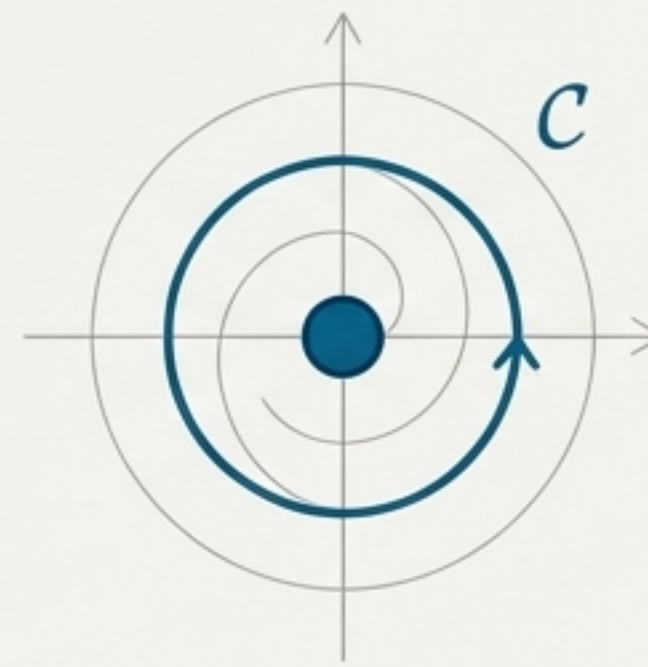
This leads directly to circulation plateaus.

$$\Gamma_C = n\kappa, n \in \mathbb{Z}$$

κ is the quantum of circulation and n is the winding number.

The Quantum of Circulation

$$\kappa \equiv \frac{h}{m_{\text{eff}}} = 2\pi r_c \|\mathbf{v}_\Theta\|$$



Loop Enclosing Core

$$\Gamma_C = \kappa$$

(for $n=1$)

Quantized Circulation Creates Quantized Time Dilation

The Implication

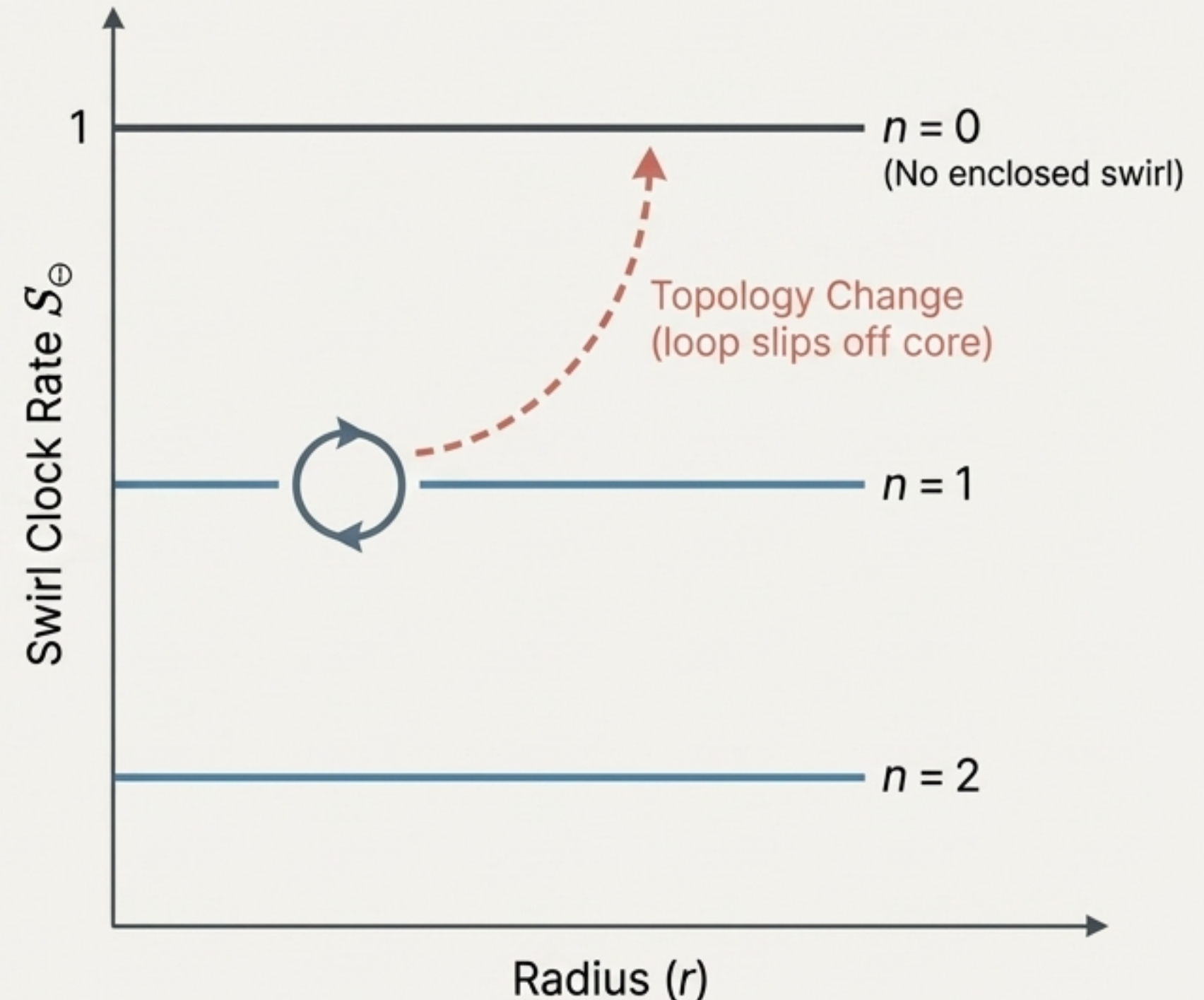
Since the tangential velocity for a circular loop is $v_\theta(r) = \frac{\Gamma C}{2\pi r} = \frac{n\kappa}{2\pi r}$, the swirl clock factor S_Θ must also be indexed by the integer n .

The Quantized Swirl Clock

$$S_\Theta(t)_n(r) = \sqrt{1 - \frac{(v_\theta^{(n)}(r))^2}{\|\mathbf{v}_\Theta\|^2}} = \sqrt{1 - \frac{n^2 \kappa^2}{4\pi^2 r^2 \|\mathbf{v}_\Theta\|^2}}$$

The Physical Meaning

The local clock rate is locked to an integer n corresponding to the loop's **topology**. Smoothly deforming a loop does not change the clock rate. A topology-changing event (e.g., a loop slipping off the core) causes the clock rate to *jump* discretely.



An Alternative View: The Swirl Lagrangian

Principle

We can derive the system's dynamics from a local, time-symmetric Lagrangian density, interpreted as a truncated form of the full non-local swirl interaction.

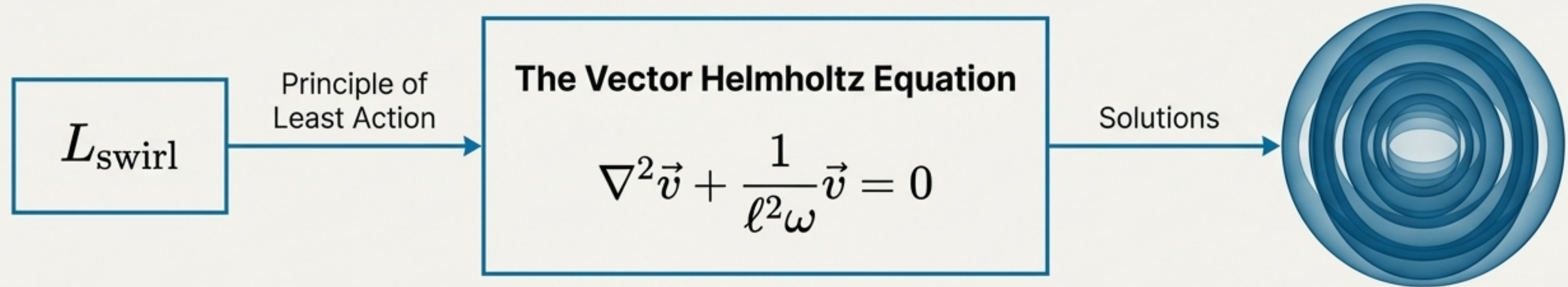
Takeaway

This formulation provides a path to derive the field equations governing stationary swirl configurations.

$$\mathcal{L}_{swirl} = \underbrace{\frac{1}{2}\rho m |\vec{v}|^2}_{\text{Kinetic Energy Term}} \underbrace{-}_{+} \underbrace{\frac{1}{2}\rho m \ell^2 \omega |\nabla \times \vec{v}|^2}_{\text{Potential Energy Term (from Swirl Tension)}}$$


The Governing Equation for Swirl Structures

Derivation: Varying the Lagrangian L_{swirl} with respect to the velocity field $\vec{v}$ and assuming an incompressible flow ($\nabla \cdot \vec{v} = 0$) yields the field equation.



- This is an eigenvalue equation for the Laplacian operator.
- Its bound solutions naturally define discrete “swirl shells” or layers of stable configuration.
- The length scale $\ell\omega$ sets the effective core radius of these solutions.

Accounting for Forces: The Swirl Stress Tensor

Purpose: To provide a complete energy-momentum bookkeeping scheme for the swirl field.

The Symmetric Stress Tensor σ_{ij}

$$\sigma_{ij} = \rho_m \mathbf{v}_i \mathbf{v}_j + \rho_m \ell^2 \boldsymbol{\omega} \left(\omega_i \omega_j - \frac{1}{2} \delta_{ij} |\vec{\omega}|^2 \right) - \delta_{ij} \frac{1}{2} \rho_m |\vec{v}|^2$$

1. **Inertial Flux.** The transport of momentum.

2. **Swirl Tension.** Represents the internal stresses due to the vorticity field.

3. **Kinetic Pressure.** An isotropic pressure-like term.

Key Analogy: Maxwell Stress Tensor

Swirl Tension (Hydrodynamic)

$$\left(\omega_i \omega_j - \frac{1}{2} \delta_{ij} |\vec{\omega}|^2 \right)$$

Direct Analogue

Magnetic Tension (Electromagnetic)

$$\left(B_i B_j - \frac{1}{2} \delta_{ij} |\vec{B}|^2 \right)$$

Synthesis: An Emergent Metric for Internal Bookkeeping

The Goal: To conveniently represent the combined effects of the swirl on internal clocks (time) and rulers (space) in a single geometric object.

$$ds^2 = -\left(1 - \frac{v_{\perp}^2}{\|v_{\Theta}\|^2}\right) dt^2 + \left(\delta_{ij} + \gamma_{ij}(\omega)\right) dx^i dx^j$$

$$g_{00} = -\left(1 - \frac{v_{\perp}^2}{\|v_{\Theta}\|^2}\right) = -S_{\Theta}^2$$



Time Dilation from
Swirl Clock S_{Θ}

$$g_{ij} = \delta_{ij} + \gamma_{ij}(\omega)$$
$$\gamma_{ij} \propto \left(\omega_i \omega_j - \frac{1}{2} \delta_{ij} |\vec{\omega}|^2\right)$$



Spatial Distortion from
Swirl Tension σ_{ij}

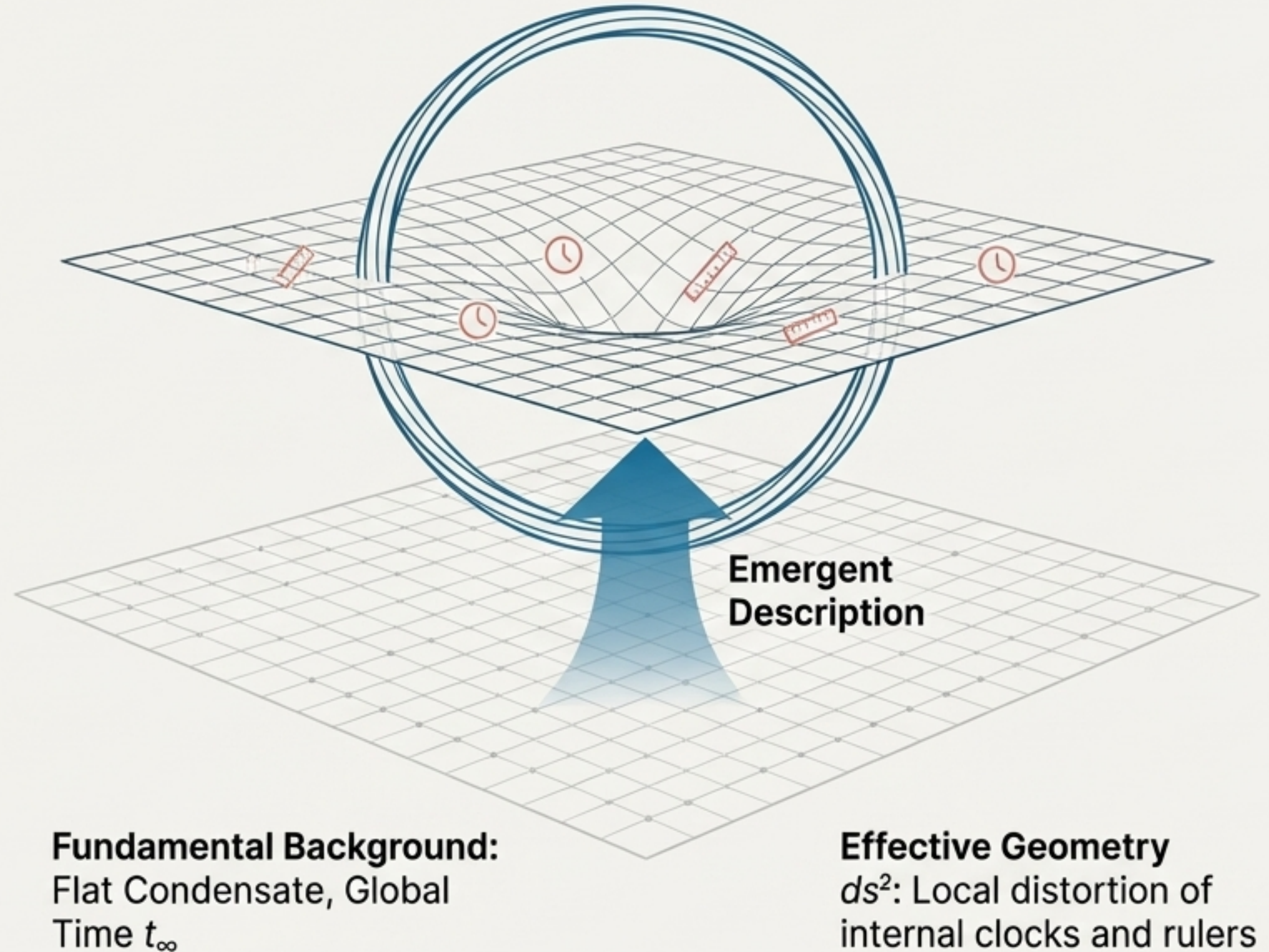
A Critical Distinction: An Effective Metric on a Flat Background

The SST Postulate

The underlying reality is a flat Euclidean spatial background and a global, absolute laboratory time, t_∞ . There is no fundamental curvature of spacetime.

The Role of the Metric

- It does not describe gravity in the sense of General Relativity.
- It efficiently summarizes how the **internal processes** of a swirl string experience distorted clocks and rulers.
- It provides an effective geometric language for a purely hydrodynamic phenomenon.



The Guiding Analogy

The Concept

A swirl string is like a spinning rubber band submerged in water.

The Connection

The faster the surface of the band spins, the faster the water around it flows. This flow, the 'swirl,' dictates the physics.

...and the faster its surface water flows, the slower a clock at its center ticks.

